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Report On Working Of AI

A Comparative Analysis of an AI Medical Diagnosis System and an Autonomous Driving System Across the PEAS Framework

This report uses the structure, terminology and comparative method of the supplied reference report, while applying the same framework to two different AI applications.

1. Theoretical Framework

1.1 The Agent–Environment Boundary

An intelligent agent can be described as a system that perceives its environment through sensors and acts upon that environment through actuators. What matters here is not what the agent is physically made of, but the relationship it maintains with its surroundings. In simple terms, the agent is part of a continuous loop: it receives information, interprets it, decides what to do, and produces an action.

The boundary between an agent and its environment is a modelling decision. In an AI medical diagnosis system, the boundary can be placed around the diagnostic model and its decision-support logic. The patient, medical devices, clinician, hospital information system and changing clinical condition are then treated as the environment. In autonomous driving, the boundary can be placed around the perception, prediction, planning and control stack, while roads, vehicles, pedestrians, traffic signals and weather belong to the environment.

This distinction helps because it tells us what information counts as a percept and what output counts as an action. It also makes the limitations of the agent clearer. A system can only make decisions using information that crosses the agent–environment boundary.

1.1 Agent–Environment Boundary — Percepts, Sensors and Actuators

A percept is one unit of sensory input at a particular moment. In medical AI, examples include a chest X-ray, a laboratory value, a reported symptom or a vital-sign measurement. In autonomous driving, examples include a camera frame, a radar detection, a lidar point cloud, a lane estimate or a GPS position.

A percept sequence is the history of information received by the agent. This distinction is important because many decisions cannot be made safely from one observation alone. A patient's previous test results can change the meaning of a new result. Likewise, the movement of a pedestrian over several frames is often more useful than one isolated image.

Sensors

Medical AI mainly receives digital information from imaging systems, laboratory systems, electronic health records and user-entered clinical information. Autonomous vehicles use physical sensors such as cameras, lidar, radar, ultrasonic sensors, GPS and inertial measurement units.

Actuators

In medical decision support, the outputs are generally informational: a risk score, diagnostic suggestion, alert, prioritisation message or recommendation to review further evidence. In autonomous driving, actuators have direct physical consequences: steering, braking, acceleration, indicators and other vehicle controls.

1.2 The PEAS Framework

Before an agent program is designed, the purpose of the agent must be specified. The PEAS framework—Performance measure, Environment, Actuators and Sensors—is a standard way of doing this. The order is meaningful: performance measure comes first because the remaining components are selected to help achieve the desired performance.

Performance measure defines what counts as success. A medical system may be judged using diagnostic sensitivity, specificity, calibration, safety, appropriate referral and response time. An autonomous driving system may be judged using collision avoidance, traffic-rule compliance, passenger safety, comfort, travel efficiency and reaction time.

Environment describes the setting in which the agent operates, including physical objects, people, rules and changing conditions. Actuators define how the agent can influence that environment, while sensors define what information the agent can obtain.

Why PEAS Matters

The four PEAS elements are connected. If a system is expected to detect a serious medical condition, it needs sensors capable of capturing relevant evidence. If a vehicle must avoid collisions, it needs sensors with enough coverage and actuators capable of responding within the required time. Skipping PEAS can therefore lead to an agent that is technically impressive but poorly matched to the actual problem.

1.2 PEAS Comparison of the Two Applications

Component	AI Medical Diagnosis	Autonomous Driving
Performance Measure	Diagnostic accuracy, sensitivity, specificity, calibration, safety, timely referral	Collision avoidance, traffic compliance, passenger safety, comfort, efficiency
Environment	Patients, clinicians, hospitals, diagnostic devices, records, changing health conditions	Roads, traffic, pedestrians, cyclists, signals, weather, road conditions
Actuators	Alerts, reports, risk scores, recommendations, case prioritisation	Steering, braking, acceleration, indicators, horn, safety alerts
Sensors	Medical images, lab results, symptoms, vital signs, patient history	Cameras, lidar, radar, GPS, IMU, ultrasonic sensors, maps

Interpretation

The medical application mainly acts through information. Its output supports a human decision-maker, especially when the consequences are significant. The autonomous driving application acts directly on a physical system, so errors can immediately affect the safety of people and objects around the vehicle.

Both systems operate with incomplete information. The difference is mainly in the type of uncertainty. Medical AI deals with biological variation and incomplete patient information, while autonomous driving deals with hidden road states and unpredictable behaviour from other road users.

1.3 The Five Agent Architectures

1.3.1 Simple Reflex Agent

A simple reflex agent selects an action using only the current percept. It can be represented as a condition–action rule: if a particular condition is observed, perform the corresponding action. Its main advantages are speed, simplicity and predictability. Its main weakness is that it cannot remember earlier information.

1.3.2 Model-Based Reflex Agent

A model-based reflex agent maintains an internal state. It updates that state as new percepts arrive, allowing it to remember useful information that is not visible at the current moment. This makes it a better fit for environments with hidden or changing state.

1.3.3 Goal-Based Agent

A goal-based agent considers a desired future state and plans actions that move toward it. It needs some representation of how actions change the environment. This makes it more flexible than a purely reactive system, although planning can require more computation.

1.3.4 Utility-Based Agent

A utility-based agent assigns preferences to possible outcomes and chooses actions that are expected to provide the best overall result. It is useful when objectives conflict, such as safety versus travel time or diagnostic confidence versus testing burden.

1.3.5 Learning Agent

A learning agent improves from experience. It can include a performance element, critic, learning element and problem generator. Learning is particularly useful when the designer cannot write reliable rules for every possible situation.

1.3.6 Summary Comparison of the Five Architectures

Architecture	Core Mechanism	Memory of Past	Plans Ahead	Handles Tradeoffs	Improves With Experience
Simple Reflex	Condition–action rules on current percept	No	No	No	No
Model-Based Reflex	Rules applied to updated internal state	Yes	No	No	No
Goal-Based	Search/planning toward a defined goal	Yes	Yes	No	No
Utility-Based	Search/planning scored by expected utility	Yes	Yes	Yes	No
Learning	Architecture refined using feedback	Yes	Depends	Depends	Yes

The table is a practical diagnostic tool. A designer should choose the least complicated architecture that supplies the capabilities required by the environment. A richer architecture is not automatically better. In a controlled environment, additional machinery can simply increase cost without improving the result.

In the following sections uses all five architectures independently for both applications. This makes it possible to see exactly what is gained when memory, planning, tradeoff handling or learning is introduced.

1.4 Dimensions of Task Environments

Task environments can be described using five main dimensions: observability, determinism, episodicity, dynamics and structure. These dimensions influence the amount of internal machinery an agent needs.

Dimension	Easier Value	Harder Value	What the Harder Value Demands
Observability	Fully Observable	Partially / Unobservable	Internal state to track what cannot currently be sensed
Determinism	Deterministic	Stochastic / Nondeterministic	Probabilistic or worst-case reasoning
Episodicity	Episodic	Sequential	Awareness of future consequences
Dynamics	Static	Dynamic / Semidynamic	Fast, time-bounded decisions
Structure	Discrete	Continuous	Approximation, interpolation or continuous control

These dimensions should not be seen as labels only. They explain why a system succeeds or fails. For example, if the environment is partially observable, memory becomes valuable. If it is stochastic, the agent needs to reason about uncertainty. If it is dynamic, decisions must be made while the world continues to change.

1.4.1 Environment Dimensions — Application Interpretation

AI Medical Diagnosis

Medical diagnosis is generally partially observable because a patient's complete physiological state cannot be directly observed. It is stochastic because the same symptom can have multiple causes. It is sequential because test results and treatment decisions affect later decisions. It is dynamic because a patient's condition can change over time. It is also partly continuous because measurements such as temperature, blood pressure and oxygen saturation vary across ranges.

Autonomous Driving

Autonomous driving is partially observable because sensors can have blind spots or noisy readings. It is stochastic because pedestrians, cyclists and drivers do not behave in perfectly predictable ways. It is sequential because a lane change or speed decision affects later positions. It is dynamic because traffic changes while the vehicle is deciding. It is continuous because speed, distance, acceleration and steering vary continuously.

The two applications therefore have similar architectural requirements even though one is a clinical information system and the other is a physical control system.

2. Case Study 1 — AI Medical Diagnosis System

2.1 Native PEAS and System Overview

The first application selected for this report is an AI medical diagnosis system. The system is considered as a decision-support agent that receives clinical evidence and produces a diagnostic suggestion, risk estimate or recommendation for further review.

Performance Measure

The system should provide useful and safe diagnostic support. Relevant measures include sensitivity for serious conditions, specificity, calibration, consistency, appropriate referral, response time and the rate of harmful false negatives. A single accuracy percentage is not enough because different errors have different consequences.

Environment

The environment includes patients, clinicians, hospitals, diagnostic devices, electronic records, laboratory systems and changing health conditions. The agent rarely sees the entire state of the patient, so uncertainty is unavoidable.

Actuators

The system may produce risk scores, alerts, diagnostic suggestions, case prioritisation and requests for additional information. In an actual deployment, these outputs support rather than silently replace professional judgement.

Sensors

Inputs may include medical images, laboratory values, symptoms, vital signs, previous diagnoses and other structured clinical information.

2.1.1 AI Medical Diagnosis — Information Flow

Stage	Input / Process	Output
1. Data Collection	Symptoms, images, laboratory values, vitals and history	Structured clinical evidence
2. Preprocessing	Cleaning, standardisation, missing-value handling	Model-ready information
3. Feature / Representation Stage	Relevant patterns extracted from data	Features or learned representations
4. Prediction	Classification or risk estimation	Probabilities / diagnostic suggestions
5. Decision Support	Compare evidence with thresholds and context	Alert, recommendation or prioritisation
6. Human Review	Clinician checks evidence and context	Final clinical decision

The workflow shows why the agent–environment boundary matters. Information can be lost or distorted before it reaches the model. Missing history, measurement errors or changes in the patient's condition can affect the final output.

Important Limitation

A high-performing model on a test dataset does not automatically become a reliable clinical agent. The deployment environment is different from a benchmark because patient populations, measurement quality, clinical workflows and consequences of mistakes are different.

2.2 The Five Environmental Profiles — Medical AI

Profile A — Controlled Diagnostic Dataset

A fixed dataset with labelled cases, standard formats and no live patient interaction. It is used for benchmarking and regression testing.

Profile B — Live Clinical Decision Support

Real patients, incomplete records, changing symptoms, multiple clinicians and time pressure. This is the closest equivalent to native deployment.

Profile C — Multidisciplinary Case Review

AI and several clinicians exchange information within a defined workflow. The protocol is structured, but the decision remains sequential.

Profile D — Emergency Triage

A high-volume stream of patients and measurements where prioritisation must be fast and uncertainty is high.

Profile E — EHR / Network Outage

The diagnostic service or record system is unavailable. The system must fall back to a safe manual procedure rather than pretending it has current information.

2.2.1 Medical AI — Profile Summary Table

Profile	Observability	Determinism	Episodic?	Static?	Discrete?
A — Controlled Dataset	Full	Deterministic	Yes	Yes	Yes
B — Live Clinical Support	Partial	Stochastic	No	No	No
C — Multidisciplinary Review	Partial / protocol-assisted	Strategic	No	Yes	Yes
D — Emergency Triage	Partial	Stochastic	Yes	No	No
E — EHR / Network Outage	None / limited	Nondeterministic	No	Yes	Yes

The profile table is intentionally similar to the reference report's approach. Profile A is a controlled environment where extra architecture is often unnecessary. Profile B is the native difficult environment. Profile C introduces structured cooperation. Profile D emphasises rapid decisions under noisy data. Profile E tests what happens when the source of information disappears.

2.3.1 Simple Reflex Agent — Medical AI

Profile	Feasibility	Key Factor
A — Controlled Dataset	High	Current case contains the information needed for the benchmark
B — Live Clinical Support	Low	No memory of patient history or changing context
C — Multidisciplinary Review	Medium	Rules can follow a defined protocol
D — Emergency Triage	Medium	Fast but sensitive to noisy measurements
E — EHR / Network Outage	High	Fixed safe fallback is appropriate

A — Controlled Dataset. PEAS: performance is based on correct classification of labelled cases. Agent program: IF a predefined symptom or image pattern matches THEN return the mapped category. Feasibility is high because the environment is deterministic, fully observable and episodic.

B — Live Clinical Support. PEAS includes safety and consistency with patient history. A reflex agent that reads only the current result can ignore earlier information and produce contradictory or incomplete suggestions. Feasibility is low.

C — Multidisciplinary Review. A rule-based agent can follow a known sequence such as requesting the next review step when a specified finding is present. Feasibility is medium because the protocol reduces uncertainty.

D — Emergency Triage. A threshold rule can quickly flag an abnormal vital sign. The weakness is that a single noisy measurement can trigger an unnecessary escalation or miss a developing problem.

E — Outage. The best behaviour is a predefined message or manual workflow. Since current records are unavailable, a richer architecture cannot recover missing information.

2.3.2 Model-Based Reflex Agent — Medical AI

Profile	Feasibility	Key Factor
A — Controlled Dataset	High	State is mostly redundant
B — Live Clinical Support	High	Patient history reduces context errors
C — Multidisciplinary Review	High	Earlier findings remain available
D — Emergency Triage	High	Rolling observations reduce noise
E — EHR / Network Outage	Medium	Cached information can become stale

A — Controlled Dataset. Internal state provides little benefit because the benchmark already exposes the relevant information.

B — Live Clinical Support. The agent maintains a patient state containing previous symptoms, results and observations. A new result is interpreted in relation to this history, making the system much more rational.

C — Multidisciplinary Review. The agent may remember which findings have already been discussed and which steps of the workflow are complete. This makes it more robust to delays or missing messages.

D — Emergency Triage. A short history of vital signs can separate a persistent trend from a single abnormal reading. This can reduce unnecessary alarms.

E — Outage. Last-known-good information can support a limited fallback, but its reliability decreases with time.

2.3.3 Goal-Based Agent — Medical AI

Profile	Feasibility	Key Factor
A — Controlled Dataset	High	Supports multi-step diagnostic reasoning
B — Live Clinical Support	Medium	Plans can be useful but patient state is uncertain
C — Multidisciplinary Review	High	Defined workflow supports planning
D — Emergency Triage	Medium	Short-horizon planning is useful
E — Outage	Low	No reliable current state for validation

A — Controlled Dataset. The agent may plan a sequence of reasoning steps to reach a diagnostic result instead of using a single pattern match.

B — Live Clinical Support. The agent may plan information gathering, such as reviewing history, examining an image and identifying whether additional evidence is needed. The plan must change when new evidence arrives.

C — Multidisciplinary Review. A goal-based system can organise evidence and decide which missing information should be considered next.

D — Emergency Triage. Short-horizon choices such as escalate, request another measurement or continue observation can be treated as goal-directed actions.

E — Outage. Planning is unsafe when the system cannot validate its assumptions against current patient information. A fixed manual fallback is preferable.

2.3.4 Utility-Based Agent — Medical AI

Profile	Feasibility	Key Factor
A — Controlled Dataset	High	Ranks multiple valid responses
B — Live Clinical Support	High	Balances confidence, safety and testing burden
C — Multidisciplinary Review	High	Balances evidence quality and time
D — Emergency Triage	High	Balances urgency and false alarms
E — Outage	Low	Current probabilities cannot be grounded

A — Controlled Dataset. Utility can rank multiple acceptable answers using correctness, confidence and clarity.

B — Live Clinical Support. The system may balance diagnostic confidence, severity, time, test burden and the cost of missing a serious condition. In practice, this is more useful than a binary rule.

C — Multidisciplinary Review. Candidate actions such as requesting another test or escalating to a specialist can be scored by expected benefit and delay.

D — Emergency Triage. Utility can balance the danger of a missed serious case against the cost of unnecessary escalation.

E — Outage. Without current patient information, utility estimates become weak and stale. A safe predefined procedure is preferable.

2.3.5 Learning Agent — Medical AI

Profile	Feasibility	Key Factor
A — Controlled Dataset	Medium	Useful for calibration and error reduction
B — Live Clinical Support	High	Validated feedback can improve future performance
C — Multidisciplinary Review	Medium	Repeated outcomes can refine ranking
D — Emergency Triage	High	Learning can improve prioritisation
E — Outage	Medium	Learning is mainly offline

A — Controlled Dataset. A critic compares predictions with known labels and the learning element adjusts the model. One important benefit is calibration and generalisation.

B — Live Clinical Support. Confirmed outcomes and carefully validated clinician feedback can be used to improve future predictions. Learning must be controlled because poor labels can teach the system the wrong pattern.

C — Multidisciplinary Review. Repeated cases can provide feedback about which evidence and recommendations were useful.

D — Emergency Triage. Learning can identify combinations of measurements associated with deterioration and improve prioritisation.

E — Outage. Past outage logs can be used to improve the offline fallback policy, but the agent cannot learn online if no live signal exists.

2.4 Medical AI — Five Architectures in One View

Architecture	What It Adds	Main Benefit	Main Weakness
Simple Reflex	Current percept → action	Very fast and easy to inspect	Ignores history
Model-Based Reflex	Internal patient state	Uses history and trends	Still reactive
Goal-Based	Explicit diagnostic goal	Can organise multi-step reasoning	Needs a reliable model
Utility-Based	Preference among outcomes	Handles risk and tradeoffs	Utility/probabilities are difficult to define
Learning	Improvement from experience	Adapts to complex patterns	Requires quality data and validation

Best-Fit Interpretation

For a realistic clinical decision-support environment, a combination of learning, internal state and utility-based reasoning is more appropriate than a simple reflex design. The final clinical responsibility should remain with appropriately trained professionals, especially for high-impact decisions.

This comparison also shows why a benchmark score should not be seen as the complete definition of an intelligent medical agent. The agent must work safely when information is missing, delayed, noisy or contradictory.

2.5 Medical AI — Risks, Reliability and Human Oversight

Data Quality

Medical models depend heavily on the quality and representativeness of their training and deployment data. Missing values, measurement errors and changes in clinical practice can reduce reliability.

Bias

If a dataset under-represents certain patient groups or conditions, performance may differ across populations. Evaluation should therefore examine subgroup performance rather than only an overall average.

Explainability and Review

A risk score is more useful when the clinician can understand what evidence contributed to the result. The system should make uncertainty visible rather than presenting a prediction as an unquestionable fact.

Human Oversight

A practical deployment keeps a trained human in the decision loop for high-impact cases. The AI system can reduce workload and highlight patterns, while the clinician provides context and final judgement.

3. Case Study 2 — Autonomous Driving System

3.1 Native PEAS and System Overview

The second application is an autonomous driving system. It is treated as an intelligent agent that continuously perceives the road, estimates the state of nearby objects, predicts likely changes, plans a safe manoeuvre and controls the vehicle.

Performance Measure

Important measures include collision avoidance, traffic-rule compliance, passenger safety, response time, ride comfort, travel efficiency and energy use. These measures can conflict. A vehicle may have to accept a small delay in order to maintain a larger safety margin.

Environment

The environment contains roads, lane markings, traffic signals, vehicles, pedestrians, cyclists, road construction, weather and changing road conditions. It is multi-agent, partially observable, stochastic, dynamic and continuous.

Actuators and Sensors

Actuators include steering, braking, acceleration and indicators. Sensors include cameras, lidar, radar, ultrasonic sensors, GPS, inertial measurement units and other vehicle-state sensors.

3.1.1 Autonomous Driving — Information Flow

Stage	Input / Process	Output
1. Sensing	Camera, lidar, radar, GPS and vehicle sensors	Raw observations
2. Perception	Detect lanes, vehicles, pedestrians, signs and obstacles	Objects and scene representation
3. State Estimation	Track objects and estimate vehicle position	Current world model
4. Prediction	Estimate likely motion of other road users	Possible future trajectories
5. Planning	Select route and safe local manoeuvre	Target trajectory
6. Control	Translate trajectory into vehicle commands	Steering, braking and acceleration

This pipeline illustrates the agent–environment loop. The world changes while the system is processing information, so the vehicle must make decisions quickly and continuously update its internal state.

Why a Single Sensor Is Not Enough

Different sensors provide different types of evidence. Cameras provide rich visual information, lidar provides geometric information, radar can provide useful distance and velocity measurements, and GPS provides global position. Combining these sources can make the agent more robust to the weakness of any one sensor.

3.2 The Five Environmental Profiles — Autonomous Driving

Profile A — Closed Simulation / Test Track

Controlled road geometry, repeatable scenarios, limited traffic and known conditions. Useful for testing and calibration.

Profile B — Urban Public Road

Real traffic, pedestrians, cyclists, weather changes, construction and imperfect observations. This is the difficult native environment.

Profile C — Cooperative Connected Traffic

Vehicles and infrastructure exchange information under a known communication protocol.

Profile D — Pedestrian and Emergency Safety

Time-sensitive situations involving pedestrians, sudden obstacles, emergency vehicles and rapidly changing conditions.

Profile E — Sensor / Network Failure

Critical sensor or network failures force the vehicle to enter a conservative fallback state such as controlled slowing or stopping.

3.2.1 Autonomous Driving — Profile Summary Table

Profile	Observability	Determinism	Episodic?	Static?	Discrete?
A — Closed Simulation	Full	Deterministic	Yes	Yes	Yes
B — Urban Public Road	Partial	Stochastic	No	No	No
C — Cooperative Traffic	Full under protocol	Strategic	No	Yes	Yes
D — Emergency Safety	Partial	Stochastic	Yes	No	No
E — Sensor / Network Failure	None / limited	Nondeterministic	No	Yes	Yes

Profile A represents a controlled testing environment. Profile B is the environment where the vehicle must deal with the full range of uncertainty. Profile C introduces cooperation. Profile D focuses on short, safety-critical episodes. Profile E examines the value and limits of architecture when reliable perception is unavailable.

3.3.1 Simple Reflex Agent — Autonomous Driving

Profile	Feasibility	Key Factor
A — Closed Simulation	High	Current sensor state can be sufficient
B — Urban Road	Low	No tracking or prediction of road users
C — Cooperative Traffic	Medium	Protocol rules can be followed
D — Emergency Safety	Medium	Fast but vulnerable to noise
E — Sensor Failure	High	Predefined fail-safe action is appropriate

A — Closed Simulation. A rule such as IF obstacle distance is below a threshold THEN brake can work well in a controlled test environment.

B — Urban Road. A reflex vehicle reacting only to the current frame cannot maintain object tracks or understand how a pedestrian or vehicle is moving over time.

C — Cooperative Traffic. A rule can follow a disclosed coordination phase, such as adjusting speed when a specific communication state is active.

D — Emergency Safety. Immediate braking rules are useful but may react incorrectly to one noisy sensor frame.

E — Sensor Failure. A fixed safe response, such as controlled slowing or stopping, is rational when critical information is unavailable.

3.3.2 Model-Based Reflex Agent — Autonomous Driving

Profile	Feasibility	Key Factor
A — Closed Simulation	High	Internal state is mostly redundant
B — Urban Road	High	Object tracking and road state improve decisions
C — Cooperative Traffic	High	Coordination state is maintained
D — Emergency Safety	High	Rolling sensor history reduces noise
E — Sensor Failure	Medium	Cached state can help briefly

A — Closed Simulation. Additional state provides little benefit because the test environment is already known.

B — Urban Road. The agent tracks nearby vehicles, pedestrians, lanes and traffic signals across time. This allows it to maintain continuity when an object is temporarily hard to observe.

C — Cooperative Traffic. The agent maintains the current communication state and recent valid messages rather than treating every message as an isolated event.

D — Emergency Safety. A short sequence of sensor observations helps distinguish a real obstacle from a temporary measurement error.

E — Sensor Failure. Last-known-good state can help initiate a safe response, but the value of that information decreases as time passes.

3.3.3 Goal-Based Agent — Autonomous Driving

Profile	Feasibility	Key Factor
A — Closed Simulation	High	Route and manoeuvre planning are predictable
B — Urban Road	Medium	The world model is approximate
C — Cooperative Traffic	High	Known protocol supports planning
D — Emergency Safety	Medium	Short-horizon planning is required
E — Sensor Failure	Low	No current observations to validate the plan

A — Closed Simulation. The vehicle can plan a sequence of turns and speed changes to reach a destination.

B — Urban Road. The system may plan a route and local manoeuvres but must replan as traffic and pedestrians change.

C — Cooperative Traffic. A known coordination protocol makes multi-step planning more reliable.

D — Emergency Safety. The system may choose among short-horizon actions such as braking, steering around an obstacle or controlled stopping.

E — Sensor Failure. A long plan is unsafe without current information. The system should instead enter a conservative fallback state.

3.3.4 Utility-Based Agent — Autonomous Driving

Profile	Feasibility	Key Factor
A — Closed Simulation	High	Ranks safe trajectories
B — Urban Road	High	Balances safety, time and comfort
C — Cooperative Traffic	High	Balances individual and shared efficiency
D — Emergency Safety	High	Balances risk and available manoeuvres
E — Sensor Failure	Low	Probabilities cannot be grounded without sensing

A — Closed Simulation. Candidate trajectories can be scored using safety margin, smoothness, time and energy.

B — Urban Road. Expected utility can balance collision risk, traffic compliance, travel time, comfort and uncertainty. This is a better fit than a single fixed rule.

C — Cooperative Traffic. The agent may consider both its own travel time and the effect of its actions on coordinated traffic.

D — Emergency Safety. The system may compare braking, steering and stopping options using predicted risk.

E — Sensor Failure. Utility estimates are unreliable when the system lacks current perception, so conservative fallback behaviour is preferred.

3.3.5 Learning Agent — Autonomous Driving

Profile	Feasibility	Key Factor
A — Closed Simulation	Medium	Useful for training and calibration
B — Urban Road	High	Learning handles varied real-world patterns
C — Cooperative Traffic	Medium	Known protocol reduces uncertainty
D — Emergency Safety	High	Experience improves rare-event handling
E — Sensor Failure	Medium	Learning improves offline fallback design

A — Closed Simulation. Repeated simulated scenarios provide data for improving perception and planning before real-world testing.

B — Urban Road. Learning can improve object detection, trajectory prediction and planning, provided the data and validation process are carefully controlled.

C — Cooperative Traffic. Learning can optimise strategy within a known communication protocol.

D — Emergency Safety. Rare events are difficult to collect, so simulation and carefully selected data can help improve the system's response.

E — Sensor Failure. Previous failure events can be used to improve fault detection and fallback transitions, but the live response should remain conservative.

3.4 Autonomous Driving — Five Architectures in One View

Architecture	What It Adds	Main Benefit	Main Weakness
Simple Reflex	Current observation → action	Fast response	No memory or prediction
Model-Based Reflex	World state estimate	Tracks objects and context	Limited future planning
Goal-Based	Explicit route / manoeuvre goals	Multi-step planning	Depends on model accuracy
Utility-Based	Preference among trajectories	Handles competing objectives	Requires reliable estimates
Learning	Experience-based improvement	Adapts to varied conditions	Needs large, representative data

The native autonomous driving environment strongly favours architectures that can maintain state, predict future behaviour and compare competing actions. A reflex layer is still useful for immediate safety rules, but it is not enough on its own for the full problem.

4. Comparative Analysis, Synthesis, and Metrics

4.1 The Knowledge Bottleneck

A central pattern appears across both applications: many failures are failures of information rather than failures of raw computation. A simple reflex medical agent may fail because it does not know the patient's earlier history. A simple driving agent may fail because the current frame does not tell it how a pedestrian has been moving.

The model-based agent addresses persistence by maintaining internal state. The goal-based agent adds knowledge of consequences and allows action sequences to be evaluated. The utility-based agent adds knowledge of relative preference, allowing tradeoffs. The learning agent adds knowledge that was not explicitly provided by the designer but can be recovered from experience.

This is why the architectures can be viewed as a progression in where useful information is stored and how it is used.

4.2 Feasibility, Complexity, and Robustness — Medical AI

Agent Type	Profile A	Profile B	Profile C	Profile D	Profile E
Simple Reflex	High / Low / High	Low / Low / Low	Medium / Low / Medium	Medium / Low / Low	High / Low / High
Model-Based Reflex	High / Medium / High	High / Medium / Medium	High / Medium / High	High / Medium / Medium	Medium / Medium / Low
Goal-Based	High / Medium / High	Medium / High / Medium	High / Medium / High	Medium / Medium / Medium	Low / Medium / Low
Utility-Based	High / Medium / High	High / High / High	High / Medium / High	High / Medium / High	Low / Medium / Low
Learning	Medium / High / High	High / High / High	Medium / High / High	High / High / High	Medium / High / Medium

Each cell is written as Feasibility / Complexity / Robustness. Complexity here is comparative rather than an exact asymptotic bound. The pattern follows the reference document: richer architectures become more valuable in uncertain live environments, while extra machinery adds little when information is unavailable.

4.3 Feasibility, Complexity, and Robustness — Autonomous Driving

Agent Type	Profile A	Profile B	Profile C	Profile D	Profile E
Simple Reflex	High / Low / High	Low / Low / Low	Medium / Low / Medium	Medium / Low / Low	High / Low / High
Model-Based Reflex	High / Medium / High	High / Medium / Medium	High / Medium / High	High / Medium / Medium	Medium / Medium / Low
Goal-Based	High / Medium / High	Medium / High / Medium	High / Medium / High	Medium / Medium / Medium	Low / Medium / Low
Utility-Based	High / Medium / High	High / High / High	High / Medium / High	High / Medium / High	Low / Medium / Low
Learning	Medium / High / High	High / High / High	Medium / High / High	High / High / High	Medium / High / Medium

The symmetry is useful: although medical diagnosis and autonomous driving operate in very different physical settings, the architectural constraints are similar once the environment is described using the same dimensions.

4.4 Comparative Synthesis Across Both Applications

Aspect	AI Medical Diagnosis	Autonomous Driving
Main uncertainty	Incomplete patient state and ambiguous clinical evidence	Incomplete road state and unpredictable road users
Most important memory	Patient history, previous tests, trends	Object tracks, lane state, recent road observations
Planning need	Sequence of diagnostic evidence and referral decisions	Route, manoeuvre and collision-avoidance planning
Utility tradeoff	Safety, confidence, testing burden and time	Safety, comfort, time, energy and legal compliance
Learning value	Pattern recognition and calibration	Perception, prediction and planning improvement
High-impact failure	Missed or misleading clinical recommendation	Collision or unsafe vehicle action
Fallback principle	Human/manual clinical procedure	Conservative vehicle safe state

The same PEAS and agent-architecture framework can therefore be used across software-based and physical AI systems. What changes is the nature of the percepts, actions and consequences.

4.5 Ethical and Social Implications

Privacy

Medical AI deals with sensitive health information, so access, storage and reuse of data require careful controls. Autonomous vehicles may collect camera and location information that can reveal people and patterns of movement.

Bias and Fairness

Medical systems can perform differently across patient groups when training data is not representative. Driving systems can also show performance differences across lighting, weather, road design and pedestrian conditions. Testing should therefore cover meaningful variation.

Accountability

An AI system should not become an excuse for unclear responsibility. Organisations need documented ownership, audit trails, testing procedures, incident review and a way to correct unsafe behaviour.

Human Role

Human judgement remains important in high-impact decisions. AI can process large amounts of information quickly, but people provide context, professional responsibility and value-based judgement.

4.6 Responsible AI and Reliability Considerations

Principle	Medical Diagnosis	Autonomous Driving
Safety	Validate clinically important errors. use escalation and review	Validate rare and common safety scenarios. conservative fallback
Transparency	Communicate uncertainty and relevant evidence	Provide understandable status and failure information
Fairness	Evaluate subgroup performance	Evaluate conditions and environments with meaningful variation
Privacy	Protect health records and images	Protect location and camera-derived information
Human Oversight	Clinician remains responsible for high-impact decisions	Human supervision and safety procedures remain important
Monitoring	Track model drift and unexpected error patterns	Track sensor faults, perception failures and unusual scenarios

Responsible AI is not a single feature added at the end of development. It affects the choice of data, performance measures, architecture, deployment process and monitoring strategy.

5. Conclusion and Future Outlook

The fifty-cell style analysis used in the reference report can be transferred to these two applications and produces a similar conclusion: rational behaviour depends on the fit between an agent's architecture and the informational structure of its environment.

Simple reflex agents are useful where the current percept contains everything needed. Model-based agents become valuable when information persists across time. Goal-based agents are useful when future consequences matter. Utility-based agents are valuable when competing objectives must be balanced. Learning agents become important when the designer cannot specify all useful patterns in advance.

The medical application demonstrates the importance of patient context, uncertainty and human oversight. The autonomous driving application demonstrates the importance of continuous sensing, prediction, planning and physical safety.

Future Outlook

Future systems are likely to combine learning, internal world models, utility-based decision-making and stronger uncertainty estimation. Medical systems may combine multiple types of clinical evidence, while autonomous vehicles may improve sensor fusion, prediction and cooperation with infrastructure.

One way to look at more general AI is not simply as a sixth architecture, but as a more capable learning and utility-oriented agent whose models and planning abilities transfer across a wider range of environments.

Appendix A: Glossary of Key Terms

Term	Definition
Agent	A system that perceives its environment through sensors and acts upon it through actuators.
Agent Program	The concrete implementation that maps percepts or percept histories to actions.
Percept	A single unit of sensory input received at one instant.
Percept Sequence	The complete history of information perceived by the agent up to a given moment.
PEAS	Performance measure, Environment, Actuators and Sensors.
Rationality	Selecting the action expected to maximise the performance measure given available information.
Simple Reflex Agent	An agent that selects actions from the current percept using condition–action rules.
Model-Based Reflex Agent	A reflex agent with internal state that tracks aspects of the environment not currently visible.
Goal-Based Agent	An agent that plans toward an explicitly represented goal.
Utility-Based Agent	An agent that selects actions by comparing expected utility.
Learning Agent	An agent whose behaviour can improve from experience and feedback.
Fully Observable	The relevant state is available to the

	agent through its sensors.
Partially Observable	Sensor information is incomplete, noisy or occluded.
Unobservable	The agent has no useful current percept information.
Deterministic	The next state is determined by current state and action.
Stochastic	The next state follows a probability distribution.
Nondeterministic	Outcomes are considered as possible alternatives without a specified probability distribution.
Episodic	Each decision episode is largely independent of other episodes.
Sequential	Current actions can affect future decisions.
Static	The environment does not change while the agent is deliberating.

Appendix A: Glossary — Continued

Term	Definition
Dynamic	The environment continues to change while the agent is deliberating.
Semidynamic	The environment does not change during deliberation, but the performance score can change.
Discrete	The environment has a finite or countable set of states, percepts and actions.
Continuous	States, percepts or actions vary over continuous ranges.
Knowledge Bottleneck	A failure caused by missing or inaccessible information needed for correct action.
Critic	The learning-agent component that evaluates performance against an external standard.
Learning Element	The component that modifies the performance element using feedback.
Problem Generator	The component that proposes exploratory actions to discover better strategies.
State	The agent's internal representation of what it believes is currently true.
Transition Model	A representation of how an action can change a state.
Performance Measure	The external criterion used to judge whether the agent is successful.
Sensor Fusion	Combining information from multiple sensors to obtain a more useful estimate.

Decision Support	AI output intended to assist a human decision-maker.
Fallback	A predefined safe behaviour used when normal operation cannot be trusted.
Model Drift	A change in the relationship between input data and desired output over time.
False Negative	A serious condition that is present but is not detected.
False Positive	A condition or event is flagged when it is not actually present.
Trajectory	A planned or predicted sequence of positions and movements over time.
Human-in-the-Loop	A design in which a person remains involved in important decisions or oversight.

Appendix B: Notation Reference, References and Student Reflection

Notation Reference

Symbol / Shorthand	Meaning
P	Set of possible individual percepts
P*	Set of all finite percept sequences
A	Set of possible actions
state(t)	Agent's internal state at time t
action(t)	Action selected at time t
percept(t)	Percept received at time t
RESULT(s, a)	State reached from state s after action a
U(s)	Utility value assigned to state s
G	Goal set used by a goal-based agent

References

1. Stuart Russell and Peter Norvig, Artificial Intelligence: A Modern Approach — concepts of intelligent agents, PEAS, task environments and agent architectures.

2. Tom M. Mitchell, Machine Learning — foundational concepts of learning systems and model evaluation.

3. Supplied reference report: 'Report On Working Of AI: A Comparative Analysis of an AI Chatbot (ChatGPT) and a Smart Home Automation System Across the PEAS Framework.' This report keeps its theoretical framework, PEAS structure, five agent architectures, environment dimensions, five profiles and 25-combination analysis for each application. The supplied document contains 25 pages and uses this exact architecture-focused organisation. □filecite□turn3file0□L705-L737□

4. General academic material on artificial intelligence, machine learning, computer vision, clinical decision support and autonomous systems.

Student Reflection

After working on these two applications, I found the PEAS framework much easier to understand because the same concepts appear in two very different situations. In the medical system, the main challenge is incomplete clinical information and the importance of human judgement. In autonomous driving, the challenge is that the physical world keeps changing while the system is making decisions.

One thing I understood from this activity is that a more complicated AI system is not automatically a better system. The architecture should match the environment and the information available to it. Memory, planning, utility and learning each solve a different limitation. This way of looking at AI makes the theory more practical and easier to connect with real applications.